

NGUYEN HOANG THACH

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Portfolio: hoangthach.me

EDUCATION

Ho Chi Minh City University of Technology and Education

Ho Chi Minh City, Vietnam

Bachelor of Engineering in Information Technology (GPA: 3.5/4.0, Expected Graduation: 2027)

Sep 2023 – Present

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Networks

PROJECTS

Agri-Supplies E-Commerce Platform

[GitHub](#)

- Developed the backend system for an agricultural e-commerce platform supporting product management, inventory, orders, and payments.
- Designed a checkout workflow with **FEFO (First Expired First Out)** inventory allocation to prioritize near-expired inventory batches.
- Applied the **State Pattern** to manage order lifecycle transitions.
- Integrated SePay for automated bank-transfer verification and implemented real-time order status updates using WebSocket.
- Implemented **Role-Based Access Control (RBAC)** for user authorization management.
- Integrated Elasticsearch for product search and built a chatbot using **Retrieval-Augmented Generation (RAG)** for product recommendations.
- Tech Stack: Java 21, Spring Boot, Spring Security, Spring Data JPA, PostgreSQL, Elasticsearch, JWT, WebSocket.*

E-Learning Platform Backend

[Live Demo](#) · [GitHub](#)

- Developed the backend system for an online learning platform supporting course management, enrollments, payments, and video processing.
- Built **RESTful APIs** using **NestJS** and **Prisma ORM** for the e-learning platform.
- Integrated **Redis** caching to improve data retrieval performance and implemented rate limiting for authentication APIs.
- Implemented JWT authentication and **Role-Based Access Control (RBAC)** for user authorization.
- Integrated Cloudinary for video uploads and media management, and implemented real-time notifications using **WebSocket**.
- Built an AI-powered dubbing pipeline using Celery and Redis for asynchronous video processing; integrated OpenAI Whisper for speech-to-text, AI translation models, and a locally hosted TTS model for voice synthesis.
- Tech Stack: NestJS, TypeScript, Prisma, PostgreSQL, Redis, Celery, Socket.io, JWT, Cloudinary.*

BlinkyVocab - Gamified Vocabulary Learning Platform

[Live Demo](#) · [GitHub](#)

- Developed the backend system for a vocabulary learning platform featuring flashcards, quizzes, and community-driven vocabulary sharing.
- Implemented a personalized learning algorithm combining **Spaced Repetition (SM-2)** and the Forgetting Curve to optimize vocabulary review schedules.
- Integrated AI services to automatically generate vocabulary metadata including definitions, examples, CEFR levels, and Vietnamese meanings from user input.
- Built features for generating vocabulary lessons by topic and extracting useful vocabulary from articles or text passages for contextual learning.
- Implemented real-time notifications using WebSocket.
- Secured the system using JWT authentication and **Role-Based Access Control (RBAC)**.
- Tech Stack: NestJS, TypeScript, Prisma, PostgreSQL, Redis, WebSockets, JWT.*

SKILLS

Programming Languages:	Java, SQL, JavaScript/TypeScript, Python
Backend:	Spring Boot, Spring Security, Node.js, Express.js, NestJS
Databases:	PostgreSQL, MySQL, MongoDB
Technologies:	Redis, Docker, WebSocket, Elasticsearch, Prisma ORM, JWT
Languages:	English (Intermediate)

ACTIVITIES & AWARDS

Winner, "Best Idea Award" - HCMUTE Hackathon 2025

2025

- Awarded for developing "TamTro" — an AI-powered mental health support platform.
- Developed backend RESTful APIs using Node.js and Express.js for authentication, real-time chat, and emotion analysis features.